



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

MECHANICAL MATHEMATICS
PAPER 1

6021/2

NOVEMBER 2019 SESSION

3 hours

Additional materials:

Answer paper

Graph paper

List of Formula MF 7

Scientific calculator

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your Name, Centre number and Candidate number in the spaces provided on the answer paper/answer booklet.

Answer **all** questions in Section A and **any five** questions from Section B.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value of g is necessary, take $g = 9,81 \text{ ms}^{-2}$

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Questions are printed in the order of their mark allocations.

The use of a scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

This question paper consists of 6 printed pages and 2 blank pages.

Copyright: Zimbabwe School Examinations Council, N2019.

Section A (40 marks)

Answer all questions in this section.

1

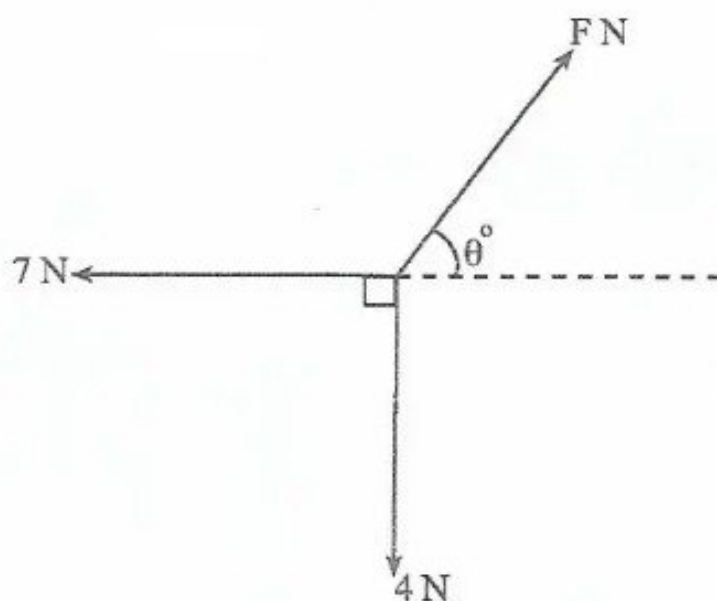


Fig.1.1

Fig.1.1 shows a particle which is in equilibrium when acted on by 3 forces as shown.

Find the values of F and θ .

[6]

2

A train of mass 30 000 kg whose engine works at a constant rate of 750 kW moves up a slope of inclination 1:50 against constant resistance of 15 000 N.

Find the

(a) acceleration, when the velocity is 7.5 ms^{-1} ,

[4]

(b) maximum speed.

[2]

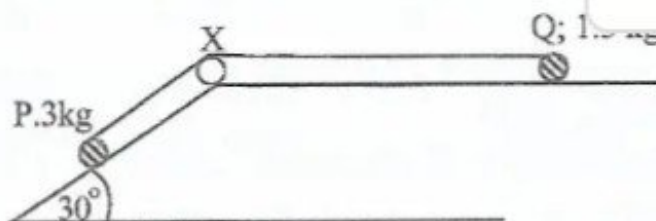


Fig. 3.1

Fig. 3.1 shows two particles P and Q held at rest on two rough planes. The particles are connected by a light inextensible string passing over a smooth pulley X fixed at the intersection of the two planes. Particle P is of mass 3 kg and the coefficient of friction between P and the plane is 0.25. Particle Q is of mass 1.5 kg and the coefficient of friction between Q and the plane is 0.50. The system is released from rest with P moving down the plane inclined at 30° to the horizontal.

Find the tension in the string and the acceleration of the particles. [8]

- 4 A ball is projected with velocity 30 ms^{-1} from a point P at the top of a cliff, 100 m above sea level, at an angle of 40° above the horizontal.

Find the

- (a) greatest height of the ball above sea level, [4]
 (b) time taken to hit the sea level. [6]

- 5 Two particles Y and Z are moving in the same direction and on the same straight line on a smooth horizontal plane. Particle Y has a mass of 0.8 kg and Particle Z has mass of 0.4 kg. Particle Y, moving with speed 7.5 ms^{-1} catches up with particle Z, whose speed is $v \text{ ms}^{-1}$. The impulse acting on Z during the collision has a magnitude of 1.6 Ns and Z moves off with a speed of 7.5 ms^{-1} .

- (a) Find the
 (i) speed v , [2]
 (ii) loss of kinetic energy. [5]
 (b) The particle Z subsequently collides and coalesces with a stationary particle of mass 0.6 kg.
 (i) Find the speed of the combined particle, after collision. [2]
 (ii) Determine whether a further collision of the combined particle with particle Y occurs. [1]

Section B (80 marks)

Answer any **five** questions in this section. Each question carries 16 marks.

6 (a)

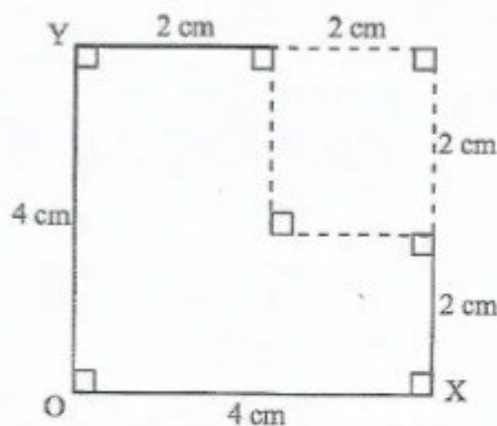


Fig. 6.1

Fig. 6.1 shows a $4\text{ cm} \times 4\text{ cm}$ square piece of card of uniform density with $2\text{ cm} \times 2\text{ cm}$ square removed from the top right hand corner.

Find the centre of mass of the remaining card.

[6]

- (b) A particle starts from rest at A and accelerates uniformly to a speed of 40 ms^{-1} in x seconds. It maintains this speed for a distance of $2\,880\text{ m}$ and then decelerates uniformly to rest at B at a rate whose magnitude is equal to twice the magnitude of the previous acceleration.

- (i) Sketch the velocity-time graph for the journey from A to B.

[4]

Given that $AB = 3\,360\text{ m}$ find the,

- (ii) value of x ,

[3]

- (ii) average speed for the journey from A to B.

[3]

- 7 (a) Two particles A and B on horizontal ground are projected vertically upwards at the same instant with initial velocities of 4 ms^{-1} and 7 ms^{-1} , respectively.

Find the

- (i) maximum height of A,

[4]

- (ii) difference between the heights of the particles when A is at this maximum.

[3]

- (iii) height of A above the ground when B is 0.5 m above A.

[6]



(b) A book falls from a shelf 240 cm above the floor.

(i) Find the speed with which the book strikes the ground. [2]

(ii) State any **one** assumption you have made. [1]

8 A particle starts from rest at point $(2;0;0)$ and moves such that at any time t seconds, the acceleration of the particle $a_{(t)}$ is given by

$$a_{(t)} = \begin{pmatrix} 16 \cos 4t \\ 8 \sin 2t \\ \sin t - 2 \sin 2t \end{pmatrix}.$$

Find the

(a) acceleration of the particle when $t = \pi$ seconds, [2]

(b) velocity of the particle when $t = \frac{\pi}{2}$ seconds, [7]

(c) displacement of the particle when $t = \frac{\pi}{4}$ seconds. [7]

9 (a) A particle is performing simple harmonic oscillations between points R and R^1 which are 10 m apart. When the particle is at a distance 3 m from the mid-point O of RR^1 , its speed is 16 m/s.

Find the

(i) period of the motion, [4]

(ii) maximum speed of the particle, [2]

(iii) maximum acceleration of the particle. [2]

(b) The displacement x , at any time t , of a particle describing simple harmonic motion is given by
 $x = 4 \cos 3t$.

(i) Find, using properties of simple harmonic motion

1. $\dot{x}$, [4]

2. $\ddot{x}$. [2]

(ii) Use your result from (i) to express $\ddot{x}$ in terms of x . [2]

- 10 A particle of mass 5 kg is being moved by a constant force of 15 N. The resistance of the particle is opposed by a resistive force of magnitude $2v$ N, where v is the particle's speed at time, t s, after passing through the origin O.



- (a) Show that $\frac{dv}{dt} = 3 - 0.4v$. [2]
- (b) (i) Given that the particle is at rest at the origin O, solve the differential equation in (a). [5]
- (ii) Hence or otherwise, find the velocity at time $t = 2$ s. [2]
- (c) (i) Express the displacement x of the particle from O in terms of t . [5]
- (ii) Hence find the displacement of the particle at $t = 2.5$ s. [2]

- 11 A particle P, of mass 1.0 kg is attached to one end of a light elastic string of natural length 0.8 m and modulus of elasticity 40 N. The other end of the string is attached to a point Q on a smooth plane inclined at 60° to the horizontal. Particle P rests in equilibrium on the plane.

- (a) Calculate the extension of the string. [3]
- (b) P is then projected down the plane from the equilibrium position with a speed of 6 ms^{-1} .

Find the extension of the string when the speed of the particle is 4 ms^{-1} for the first time. [13]

12

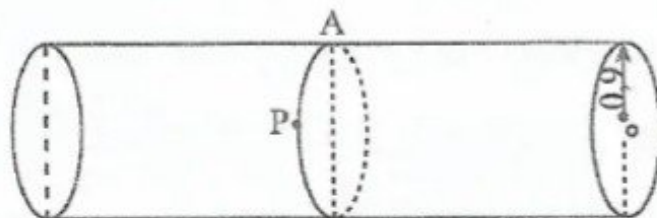


Fig. 12.1

A smooth cylinder of radius 0.9 m is fixed with its axis horizontal on a horizontal plane. A small bead of mass 0.03 kg is placed at the highest point A, on the outside of the cylinder and just displaced from rest. (see Fig. 12.1) The bead subsequently loses contact with the surface of the cylinder at point P.

- (a) Find the height of P above the axes of the cylinder. [9]
- (b) Calculate the angle between the horizontal and the direction of motion of the bead when it leaves the cylinder. [1]
- (c) Find the speed and tangential acceleration of the bead at P. [4]
- (d) Describe the subsequent motion of the bead. [2]